



Regular Article

Who gets to use ChatGPT? A global study on digital access and inequality in higher education[☆]Andrea Bencsik^a , Szonja Jenei^{b,*} , Szilvia Módosné Szalai^b ^a Wekerle International University, Hungary^b Széchenyi István University, Hungary

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ABSTRACT

This study examines how national-level digital development – measured by the ICT Development Index (IDI) – affects university students' use of ChatGPT. Special emphasis is placed on mediating factors that may influence this relationship, including technical access, institutional and linguistic support, and individual background characteristics, particularly in relation to educational equity and sustainability (SDG 4).

The analysis is based on survey data from 20,242 students across 58 countries and applies multivariate statistical methods, including logistic regression, PLS-SEM modeling, and cluster analysis. The results indicate that students in countries with higher IDI scores are more likely to use ChatGPT, primarily because of more advanced digital competencies and greater technological access. The country of study proved to be a stronger predictor than citizenship, underscoring the key role of the local educational environment.

Functional access emerged as the most decisive mediating factor, while institutional and linguistic support had a more indirect effect on usage. Cluster analysis identified three distinct student profiles and highlighted that a high level of digital infrastructure alone does not ensure the widespread adoption of generative AI tools.

The study proposes a multi-level interpretive framework: at the macro level, national digital infrastructure; at the meso level, institutional and linguistic support; and at the micro level, individual characteristics – connected by functional access as a mediating dimension. This context-sensitive approach contributes to a more comprehensive and practice-oriented understanding of digital inequalities and the integration of generative AI in higher education, offering guidance for promoting inclusive and sustainable technology use.

1. Introduction

Generative language models – such as ChatGPT – represent one of the most significant technological innovations of recent years. Their capabilities in generating natural language texts, interpreting complex information, and engaging in dialogue are fundamentally reshaping not only education but also other knowledge-intensive sectors, such as healthcare and public administration (Sharma & Ha, 2025). At the same time, this technological advancement amplifies existing inequalities in digital access, particularly in regions with weak infrastructure and low levels of digital competence (Nedungadi, Tang & Raman, 2024). Van Dijk's (2005) digital divide theory – discussed in detail in Section 2.1 – captures these disparities along three key dimensions: socio-economic status, national digital development, and the institutional environment

(Aad & Hardey, 2025).

To measure technological differences between countries, the International Telecommunication Union (ITU) developed the ICT Development Index (IDI), which is introduced in detail in Section 2.2. While the IDI effectively reflects national-level digital capacities, only a limited number of studies have examined how it influences the use of artificial intelligence tools for educational purposes. The integration of generative AI depends not only on individual digital proficiency but also on a variety of structural factors – including access to linguistic support, institutional policies, and the quality of digital infrastructure. Although the related literature is expanding rapidly, there remains a lack of globally comparable data on how national-level digital development affects students' use of AI tools, their attitudes, and learning experiences. Sharma and Ha (2025), for example, demonstrated in their

^{☆ 1} the ICT Development Index (IDI), published annually by the International Telecommunication Union, which serves as a composite indicator of digital infrastructure and usage.

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comparative study of students in the United States and Bangladesh that the use of generative AI is shaped not only by technical competencies but also by the broader social and infrastructural context.

This study examines the integration of generative AI in higher education across four interrelated levels. The macro level encompasses national digital development and infrastructure (IDI); the meso level includes institutional and linguistic support structures; and the micro level focuses on individual demographic and academic characteristics. Positioned between these is the fourth, mediating level: functional access. This concept appears not only as a variable, but also as an analytical level that helps uncover how structural opportunities are transformed into actual tool usage. This multi-level framework is particularly well-suited for investigating the conditions under which generative AI can contribute to inclusive learning – and the circumstances under which it may risk exacerbating existing educational inequalities.

The theoretical background of the analysis integrates multiple approaches: Van Dijk's (2005) digital divide theory, the knowledge management perspective, the UTAUT model by Venkatesh et al. (2003), and the theoretical framework of socio-technical systems. This interdisciplinary foundation enables a complex, multi-level interpretation of technology use and digital inequalities. The study offers a novel scientific contribution in three key respects. First, it applies a comprehensive, multi-level theoretical model that incorporates not only individual-level but also macro- and meso-level factors in interpreting the integration of generative AI in higher education. Second, it employs a methodologically robust research approach that combines cluster analysis with PLS-SEM modeling, allowing for the simultaneous exploration of structural and contextual effects. Third, it draws on the dataset from the *Students' Perception of ChatGPT* study, which includes responses from 23, 218 students across 109 countries – thereby enabling a globally relevant, empirical investigation of the relationship between digital inequalities and artificial intelligence in the context of higher education.

The structure of the study is as follows: Section 2 reviews the relevant literature and the applied theoretical frameworks, Section 3 presents the research methodology, Section 4 reports the results, Section 5 discusses the theoretical and practical implications, and Section 6 summarizes the key findings and offers suggestions for future research directions.

2. Literature review

This section reviews the theoretical frameworks and empirical findings that form the foundation of the study. The focus is on the factors influencing the use of generative artificial intelligence – particularly ChatGPT – in higher education. The analysis is structured around three interrelated levels: the macro level, which includes national digital infrastructure and development; the meso level, which covers institutional and linguistic support; and the micro level, which addresses students' demographic characteristics and academic background. *Functional access* plays a mediating role between these levels.

The applied theoretical models make it possible to explore how factors operating at different levels influence the use of generative AI tools for academic purposes. Particular emphasis is placed on the reliability of internet access, the availability of digital devices, access to digital services, the forms of institutional support, and students' digital literacy and AI-related knowledge. These theoretical foundations serve as the basis for formulating the research questions and the analytical model presented in Section 2.6.

2.1. Theoretical perspectives on AI use in education

The study is grounded in four complementary theoretical frameworks: the digital inequality model (Van Dijk, 2005), knowledge management theory (Rodríguez-Pasquín & López-Martínez, 2024), the UTAUT model (Venkatesh et al., 2003), and the socio-technical systems approach (Stewart et al., 2024). Together, these frameworks enable a

comprehensive interpretation of AI use in higher education, with particular attention to social background, institutional conditions, and individual digital competencies. The theory of digital inequality focuses on disparities in technological access, digital skills, and usage patterns – dimensions that have recently been extended by factors such as algorithmic awareness and AI literacy (Aad & Hardey, 2025). Knowledge management theory conceptualizes AI as a tool for knowledge creation and sharing, with its effectiveness depending on students' digital competence and the degree of institutional support.

The UTAUT model (Venkatesh et al., 2003) identifies four core determinants – performance expectancy, effort expectancy, social influence, and facilitating conditions – that shape individuals' intention to use AI; these are also influenced by demographic characteristics (Adjekum & Asare, 2025). The socio-technical perspective views AI implementation as the outcome of interactions among organizational, pedagogical, and societal factors that collectively enable effective, context-sensitive use of technology (Stewart et al., 2024). Research associated with this framework emphasizes the importance of ensuring equitable access, skills development, and ethical use (Duong et al., 2024). ICT development – as an indicator of digital infrastructure and institutional capacity – plays a key role in determining how and to what extent students use AI tools (Yakubu, 2025). According to Sharma and Ha (2025), social and technological conditions vary across countries and influence the adoption of generative AI tools in distinct ways.

2.2. National ICT readiness and structural conditions

Teachers' attitudes toward the educational use of artificial intelligence have become a central focus of recent research. The UTAUT model (Venkatesh et al., 2003) identifies four factors – expected performance gains, usability, social influence, and facilitating conditions – that shape educators' intentions to use technology. According to international studies, teachers are more likely to adopt generative AI tools when clear guidelines, technical support, and professional development opportunities are available. Tram (2025) argues that language teachers' attitudes are primarily influenced by their pedagogical preparedness and their perceptions of the technology's usability. According to Barakat et al. (2025), the acceptance of ChatGPT is shaped by perceived usefulness and its compatibility with pedagogical practice. Overall, research suggests that sustainable adoption of generative AI is feasible only if the institutional and pedagogical environments align with teachers' professional identity and support needs.

2.3. Student characteristics and education

While digital infrastructure is a fundamental prerequisite for technological access, the use of generative AI is primarily shaped by students' socio-demographic characteristics and learning environments. Gender, age, socio-economic background, and prior digital experiences significantly influence students' confidence in using AI tools, their trust in the technology, and the frequency of use (Huang et al., 2025). Gender and socio-economic disparities shape student experiences: female students and those from lower-income backgrounds are more likely to experience ethical uncertainty or anxiety, especially in the absence of clear institutional guidance. In contrast, students with advanced digital skills and strong self-regulated learning competencies tend to use AI tools more consciously and critically (Weng et al., 2024). The mode of learning also plays a role. Students participating in distance or hybrid education tend to exercise greater autonomy and use AI tools more intensively than those in traditional face-to-face settings (Banihashem et al., 2025). Disciplinary differences are also observed. AI integration is more common in technical, IT, and business programs, while it is less frequent in the humanities and arts – partly due to differences in curricula and pedagogical approaches (Chiu, 2024).

Motivation is another influential factor. Students with intrinsic, development-oriented goals tend to view AI as a supportive tool. In

contrast, those driven by extrinsic motivation, such as grade attainment, are more likely to engage in superficial use (Anders & Speltz, 2025). Conscious and responsible use is promoted by clear regulations, transparent assessment frameworks, and institutional programs aimed at developing digital competence (Shi et al., 2025). Research suggests that students' characteristics, educational backgrounds, and learning environments interact to shape AI usage patterns, potentially leading to new forms of learning inequality. Even among students with similar demographic profiles, significant differences may arise due to variations in infrastructure, linguistic accessibility, or institutional support.

2.4. Barriers and enablers of equitable AI use

Even in digitally advanced environments, the use of generative AI tools is shaped by intermediary factors that determine whether technological access genuinely supports equitable learning. This study identifies three key factors: functional access, linguistic accessibility, and institutional support – each influencing students' opportunities and limitations at different levels. *Functional access* refers not only to the availability of devices or internet connection, but also to their stable, reliable, and learning-conducive usability. Poor connectivity, device sharing, or lack of time – for instance, due to employment or family obligations – can hinder independent learning with AI tools and ultimately impede skill development (Fan, 2025).

Linguistic accessibility presents a significant barrier, particularly for students whose first language is not English. While AI tools are technically global, the dominance of English, along with cultural references and linguistic nuances, complicates effective use. With lower language proficiency, it becomes difficult to formulate accurate prompts and critically interpret the generated content – especially in the absence of multilingual interfaces and support materials (Chan & Colloton, 2024). *Institutional support* – including clear policies, training opportunities, guidelines, and advisory services – is also essential. In its absence, distrust and uncertainty may emerge; by contrast, consistent communication and a supportive environment facilitate conscious and inclusive technology use (Jin et al., 2025). The literature thus suggests that formal access alone is insufficient – meaningful learning experiences can occur only when students possess the skills needed for the conscious and equitable use of AI tools.

2.5. Digital competence, inclusion, and ethical use

The skills dimension of the ICT Development Index (IDI) reflects the educational policy priorities of individual countries and the extent to which they emphasize the development of digital competence (AlSagri & Sohail, 2024). This section explores how students' digital preparedness and perceived institutional support influence their access to AI tools and their conscious, responsible use. Digital competence encompasses technical, cognitive, and ethical skills. In the context of generative AI, this includes formulating clear prompts, critically evaluating content, verifying sources, and identifying bias (Alamäki et al., 2024). These skills are essential to ensure that AI functions not merely as a tool, but as a value-driven solution that meaningfully supports learning. Digital competence is closely linked to AI literacy, which refers to the ability to interpret AI-generated content consciously, analytically, and ethically. While the two concepts can be distinguished, in practice they often appear together – especially when students engage with AI tools independently, without institutional guidance (Torres-Rivera et al., 2025). Technological access alone is insufficient for effective learning – it must be complemented by appropriate technical, cognitive, and ethical competences (Nedungadi, Tang & Raman, 2024). Developing digital competence is not solely an individual responsibility; it is also an institutional and societal obligation. Nevertheless, educational systems frequently treat it as a “soft skill” and delegate its development to individuals, thereby increasing inequalities (Flores-Viva & García-Peñalvo, 2023). For disadvantaged students, paywalled, linguistically

limited, or culturally irrelevant learning materials may pose significant barriers.

Many students also report experiencing ethical uncertainty – they are unclear about the appropriate purposes and acceptable extent of AI use. A lack of regulation and instructor guidance can lead to anxiety, which hinders responsible and creative engagement. The sustainable educational integration of generative AI can only be achieved if it aligns with the United Nations Sustainable Development Goal 4, which emphasizes equitable access to quality education. Achieving this requires coordinated efforts to develop digital competence and reduce educational inequalities. Collaborative learning formats, peer knowledge sharing, and the use of multilingual instructional materials may support more equitable and conscious use of AI tools (Apata et al., 2025).

2.6. Research questions

This study investigates the use of generative artificial intelligence in higher education, based on the theoretical frameworks presented in the previous sections – including digital inequality, knowledge management, the UTAUT model, and socio-technical systems. The aim is to explore the contextual, institutional, and individual factors that influence the acceptance and application of AI tools in education. According to recent literature, technological availability alone does not ensure the effective use of AI tools. Critical factors include the level of infrastructure development, the systemic nature of institutional support (Rochmah, 2025), and students' digital competence and AI literacy (Hoshino & Siu, 2025). The research questions are designed to explore how structural, mediating, and background factors shape the global diffusion and use of AI tools in higher education. The following research questions span multiple levels of analysis – national, institutional, and individual – and serve as the foundation for the analytical model and hypothesis framework presented in Section 3.

RQ1. To what extent does students' citizenship influence their use of ChatGPT in higher education?

Citizenship defines a legal and institutional framework that influences access to digital technologies, the availability of educational services, and the accessibility of linguistically adapted learning materials. Cultural attitudes, openness to innovation, and trust in regulation may also shape patterns of artificial intelligence use (Hoshino & Siu, 2025).

RQ2. To what extent does students' country of origin affect their use of ChatGPT in higher education?

Country of origin influences students' early digital experiences and shapes their attitudes toward AI tools and academic integrity norms. The country's level of socio-economic development, the extent of ICT investment, and the quality of language education all affect how confidently and responsibly students are able to use AI tools – even when they are currently studying in a different country (Rochmah, 2025).

RQ3. To what extent does the country where students pursue their studies influence their use of ChatGPT?

The specific characteristics of the country of study – such as the language of instruction, institutional attitudes toward AI tools, and relevant regulations – may independently influence the adoption of AI in higher education (Huang et al., 2025). This effect may occur regardless of students' citizenship or country of origin.

RQ4. How is the level of national ICT development (ICT Development Index) associated with students' use of ChatGPT in higher education?

The development of digital infrastructure is an important starting point, but it does not in itself guarantee the conscious and effective use of generative AI tools in education. The IDI score becomes a true advantage only when key mediating factors are also in place – such as stable access, linguistic support, and clear institutional guidelines (Adjekum & Asare, 2025).

RQ5. How are students' demographic and academic characteristics – such as gender, financial status, field of study, mode of learning, age, student status, level of study, and employment – related to their use of ChatGPT?

Students' demographic and academic characteristics – such as gender, age, field of study, income level, or employment status – significantly influence how frequently and for what purposes they use AI tools. Among students in STEM programs, AI is more commonly used in a productive, learning-oriented manner, whereas in other disciplines, ethical concerns or creative uncertainty tend to arise more frequently (Zhang & Wang, 2025).

RQ6. How do functional access, language support, and institutional support mediate the relationship between national ICT development and students' use of ChatGPT?

Technological advancement alone is not sufficient for the meaningful and responsible use of generative AI tools – stable access, linguistic support, and a supportive institutional environment are also essential. Training programs grounded in AI ethics and instructor mentoring play a key role in helping students feel safe and in fostering positive attitudes toward these tools (Duong et al., 2024).

RQ7. Does national ICT development have a direct effect on ChatGPT usage that is not fully explained by mediating factors such as access and support?

ICT development reflects not only technical capacities but also cultural and educational policy patterns, which can directly influence the integration of AI tools in education. This helps explain why the practice of AI use may differ even among countries with similar technological resources (Huang et al., 2025). The research questions are embedded in a multi-level analytical model that enables the examination of generative AI in higher education in relation to global digital inequalities and institutional conditions. The following section presents the methodological approach, including details on data collection, sampling, and analytical procedures.

3. Methodology

The study aims to explore the extent to which national-level digital development, as measured by the ICT Development Index (IDI), influences university students' use of ChatGPT. The analysis considers not only direct effects, but also how this relationship is mediated by functional access, institutional support, and linguistic barriers. In addition, the study accounts for students' demographic and academic characteristics and compares the relative impact of country of origin, citizenship, and country of study.

The investigation is based on an international, publicly available, and anonymously collected dataset organized by the CovidSocLab research group at the Faculty of Public Administration, University of Ljubljana, under the leadership of Prof. Dr. Aleksander Aristovnik. Data collection took place between October 2023 and February 2024, with contributions from nearly 150 international researchers. One of the current study's co-authors, Prof. Dr. Andrea Bencsik, joined the scientific analysis of the project after the data collection phase.

The original dataset contained 23,218 student responses from 109 countries. After data cleaning, 20,242 valid responses from 58 countries

– each with at least 35 participants – were included in the final analysis. The dataset and the questionnaire are publicly accessible: Ravšelj et al. (2025), *Higher Education Students' Early Perceptions of ChatGPT: Global Survey Data*. Mendeley Data. <https://doi.org/10.17632/ymg9nsn6kn.2>.

The questionnaire consisted of eleven sections and provided a comprehensive examination of higher education students' perceptions, usage patterns, and attitudes related to ChatGPT, as well as their educational and social backgrounds. The first section captured respondents' sociodemographic characteristics, followed by questions assessing the frequency, purposes, and user experience of ChatGPT usage. The survey addressed perceived technological capabilities, attitudes toward ethical and regulatory issues, and overall user satisfaction.

Subsequent sections focused on academic performance, skills development, labor market expectations, and emotional responses associated with the use of the tool. The final sections explored students' educational background, economic situation, living environment, personality traits, and general opinions about ChatGPT, also allowing for open-ended responses. The geographical and demographic diversity of the sample enables reliable global comparisons of generative AI use in higher education, as well as robust analysis of the relationship between national ICT development and technology usage.

3.1. Analytical strategy

The analysis was conducted in three consecutive stages, following both inductive and deductive methodological logic. In the first stage, binary logistic regression was used to examine the extent to which the likelihood of using ChatGPT was influenced by structural factors (e.g. country of origin and country of study) and individual factors (e.g. gender, financial situation, mode of study). All analyses were performed using IBM SPSS Statistics 29.0 (Hosmer et al., 2013). In the second stage, student usage patterns were explored through a two-step cluster analysis. The SPSS TwoStep algorithm determined the optimal number of clusters based on the Bayesian Information Criterion (BIC), which was further refined using the K-means procedure.

In the third stage, the effects of explanatory variables were further analyzed through additional logistic regression (ENTER method with stepwise model building), followed by structural equation modeling (PLS-SEM) to uncover deeper relationships. These analyses were conducted using SmartPLS 4.0. The model presented in Fig. 1 illustrates the core assumption of the study: students' use of generative AI tools is shaped not only by individual decisions but also by the level of digital infrastructure in their country, as well as their demographic and academic characteristics. Since the IDI – a national-level variable – was assigned to individual respondents based on their citizenship, country of origin, or country of study, there is a risk of ecological bias. Although the IDI provides important structural context, it does not reflect intra-country disparities in access among respondents. To mitigate this bias, separate model estimations were conducted based on different country identifiers, and macro-level indicators were not interpreted as direct proxies for individual capabilities or access. Furthermore, demographic control variables were intentionally excluded from the cluster analysis to avoid additional bias arising from level-mixing between individual- and country-level data.

The next section elaborates on this conceptual model, with special attention to the mediating mechanisms through which the digital environment shapes ChatGPT usage patterns.

3.2. Research model

The use of ChatGPT in higher education was examined through a theoretically grounded, multi-level research model (Fig. 2). The model is based on the assumption that students' use of AI tools is determined by three groups of factors:

- national digital development (macro level),

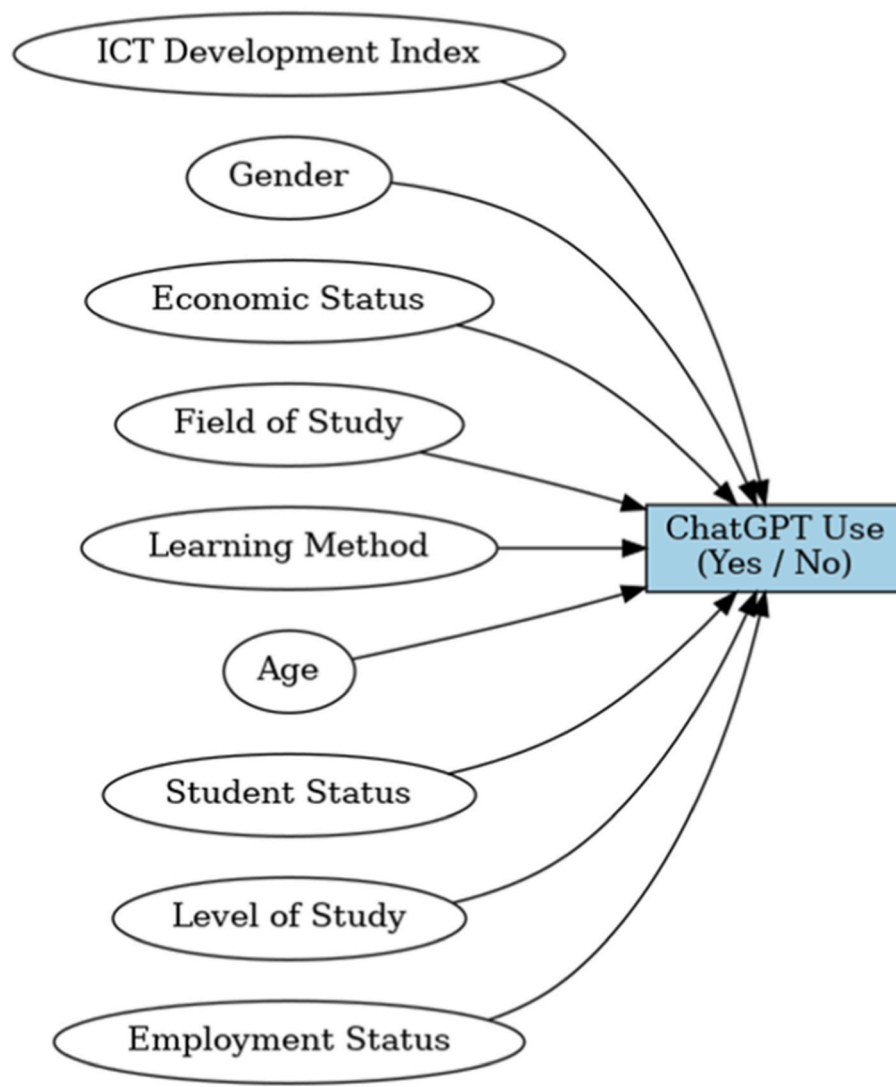


Fig. 1. Factors Influencing ChatGPT Use in Higher Education. Source: Authors' elaboration.

- institutional and linguistic support (meso level), and
- individual demographic and academic characteristics (micro level).

To test the hypotheses, we applied partial least squares structural equation modeling (PLS-SEM), a method well suited for analyzing complex, multi-level models with heterogeneous samples (Hair et al., 2021). This approach enabled the simultaneous analysis of both direct and indirect effects, as well as formative and reflective constructs – which was essential for the multi-level design of the study (Hair et al., 2019).

ChatGPT use was included in the model as a binary variable (0 = did not use, 1 = used), and this coding was applied consistently across the entire dataset. The ICT Development Index (IDI) was treated as a continuous, interval-scaled variable. The reliability of multi-item constructs was assessed using Cronbach's alpha, composite reliability (CR), average variance extracted (AVE), and the heterotrait-monotrait ratio (HTMT). These reliability indicators were not applicable to single-item variables such as IDI and ChatGPT use.

For the formative ICT Development Index (IDI) construct, multicollinearity was assessed using variance inflation factors (VIF). In the structural model, we examined path coefficients, explained variance (R^2), predictive relevance (Q^2), and the significance of effects using a 5000-sample bootstrap procedure (Hair et al., 2019). On the left side of

the model, the IDI appears as a formative macro-level construct representing the level of national digital infrastructure and competences (ITU, 2024). The index is based on objective statistical indicators – such as broadband coverage, mobile access, and levels of digital skills – and draws on Helsper's (2021) digital inequality ecosystem model. The relationship between the IDI and ChatGPT use is mediated by three reflective constructs:

- *Functional access*, which captures the frequency and purpose of integrating generative AI tools into academic activities (Q15, Q18a–k).
- *Linguistic support*, which reflects perceived language-related benefits, such as improvements in writing and communication skills (Q28d, Q28f).
- *Institutional support*, which assesses the awareness and acceptance of guidelines and ethical norms regulating AI use (Q21c, Q23a–c).

These constructs help to explain how the advantages stemming from digital infrastructure are transformed into actual learning access. Previous research has also emphasized that technological availability alone is insufficient; the presence of linguistic and institutional support plays a critical role in the effective academic integration of AI tools (Long & Magerko, 2020). The model aims to examine the extent to which the IDI influences the likelihood of using ChatGPT for academic purposes, both directly and indirectly. This approach helps to uncover how macro-level

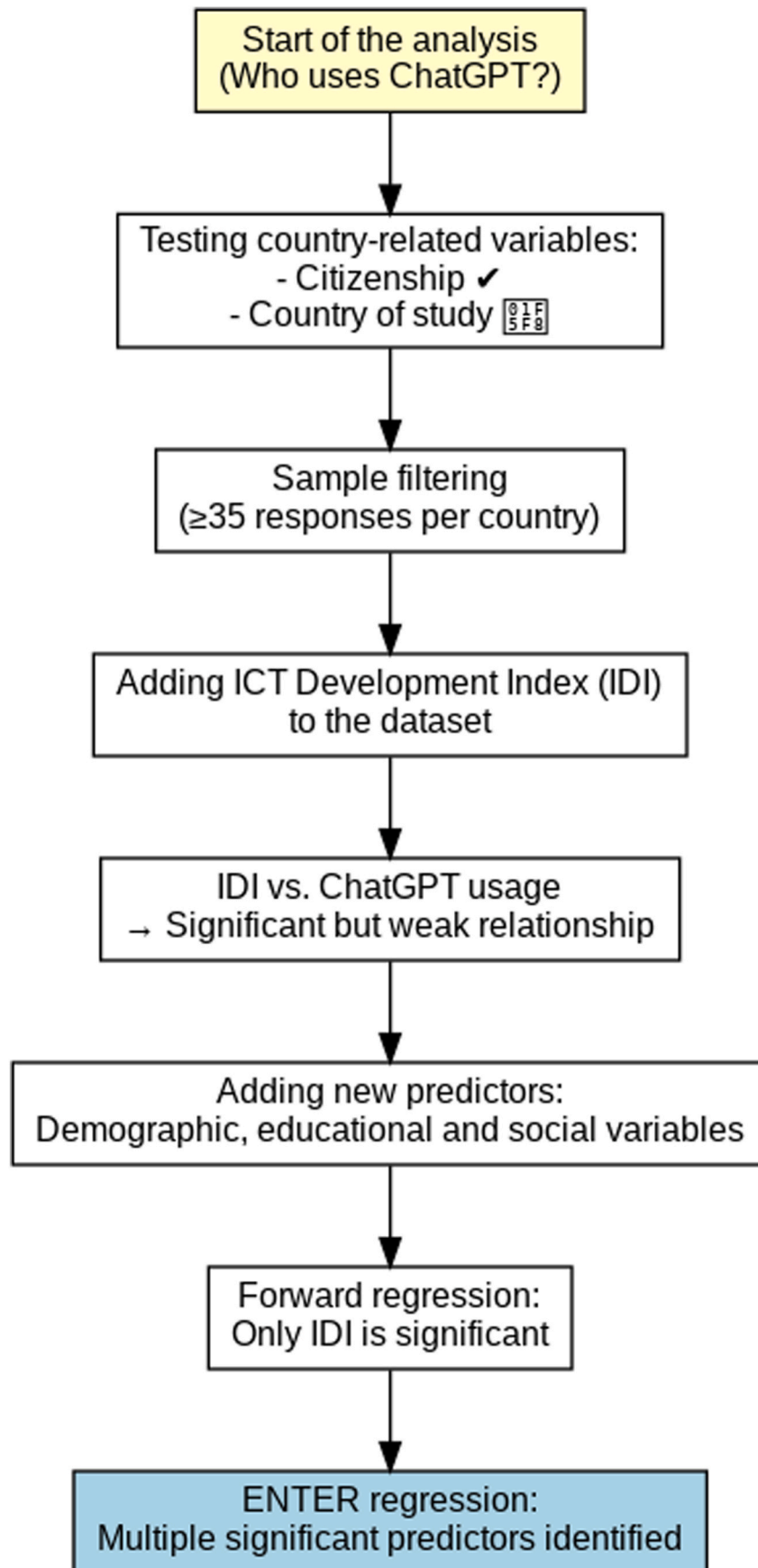


Fig. 2. Research Model. Source: Authors' elaboration.

digital inequalities shape individual learning opportunities in the context of generative AI use in higher education.

3.3. Cluster analysis

In the final stage of the analysis, cluster analysis was used to explore the patterns of access that characterize students' use of generative AI tools. Our goal was not to identify social or demographic types, but rather to detect distinct usage and contextual profiles, with particular attention to the combination of functional access, institutional and linguistic support, and national digital development. The hybrid clustering procedure included the average scores of the three reflective constructs – functional access (ACCESS), linguistic support (LANGUAGE_SUPPORT), and institutional support (SUPPORT) – along with country-level IDI values as input variables.

The optimal number of clusters was determined using the SPSS TwoStep algorithm based on the Bayesian Information Criterion (BIC), and cluster boundaries were refined using the K-means method. Demographic variables were deliberately excluded from the clustering process, as our objective was to form groups based on usage and contextual patterns, rather than social categories. In addition, demographic variables were not consistently available across all countries and were therefore analyzed separately using logistic regressions.

To assess clustering effects, we ran an empty random-intercepts model at the country level. The intraclass correlation coefficient (ICC = .081) indicated a moderate level of between-country variance, suggesting that while the country-level clustering effect is present, it is not dominant. Accordingly, individual-level analysis can be considered valid and unbiased.

The quality of the clustering solution was evaluated using the Silhouette coefficient, which yielded a value of $S = .42$. Since a Silhouette score above 0.25 indicates acceptable cluster separation, the cluster structure can be considered statistically robust, and additional ANOVA testing was deemed unnecessary. The cluster analysis is related to Research Questions 3 and 4 (RQ3 and RQ4), which focus on the patterns of ChatGPT use and the factors influencing them (e.g. functional access, digital competence, institutional trust). Its theoretical foundation lies in the second level of Van Dijk's (2020) digital divide theory, which emphasizes differences beyond mere technological access – particularly in how technologies are used and the extent to which they are beneficial. The practical significance of the identified clusters lies in their potential to inform targeted policy interventions. Our findings may assist higher education institutions and AI platform developers in adapting their systems and guidelines to better meet the diverse needs of students (Long & Magerko, 2020).

3.4. Hypotheses

The following hypotheses are based on the research questions presented in Section 2.6, and aim to test the proposed multi-level theoretical framework empirically. The model conceptualizes students' use of ChatGPT as the combined effect of macro-level digital development, meso-level institutional and linguistic factors, and micro-level demographic and academic characteristics.

RQ1 examines the role of citizenship, which is considered a structural background variable reflecting students' digital, educational, and cultural environments (Ful Haq et al., 2025). These structural conditions typically facilitate or constrain the adoption of emerging technologies, making the expected direction of influence theoretically definable. Based on this rationale, we propose the following: **H1: Students with more digitally and culturally advanced citizenship backgrounds are more likely to use ChatGPT than those from less supportive environments.**

RQ3 focuses on the country of study, which directly shapes the language of instruction, the degree of institutional digitalization, and access to technology (Bukhari, 2025). Since these factors may exert influences in different directions, the theoretical relationship is not

unidirectional. Taking the country of study into account, we propose the following hypothesis: **H2: The country in which a student is pursuing their studies significantly influences the likelihood of ChatGPT use.**

RQ4 investigates the role of national digital development, as measured by the ICT Development Index (IDI). Higher IDI values are associated with more advanced infrastructure, greater digital competence, and wider technological diffusion (Ful Haq et al., 2025), all of which are typically linked to increased technology adoption. Based on the expected impact of national digital development, we hypothesize: **H3: Students from countries with higher ICT Development Index scores are more likely to use ChatGPT.**

RQ5 examines the influence of students' demographic and academic characteristics. Prior empirical research suggests that these factors generally affect technology use systematically and directionally (Zhang, 2024). In light of the known mechanisms by which individual demographic and academic traits influence behavior, we propose the following: **H4: Students' demographic and academic characteristics systematically influence the likelihood of ChatGPT use, either positively or negatively.**

Sub-hypotheses:

- **H4a:** Male students are more likely to use ChatGPT than female students.
- **H4b:** Students with a stronger financial background are more likely to use ChatGPT.
- **H4c:** Students enrolled in digitally intensive fields of study are more likely to use ChatGPT.
- **H4d:** Students in online or hybrid learning formats are more likely to use ChatGPT than those in traditional in-person programs.
- **H4e:** Younger students are more likely to use ChatGPT than older students.
- **H4f:** Full-time students are more likely to use ChatGPT than part-time students.
- **H4g:** Students at higher levels of education (MA/PhD) are more likely to use ChatGPT.
- **H4h:** Students with work experience are more likely to use ChatGPT.

RQ6 investigates the mechanisms through which national digital development indirectly affects ChatGPT use. According to the literature, higher levels of digital development enhance technological access, linguistic support, and institutional capacity (Ful Haq et al., 2025), all of which positively influence the use of AI tools. Based on theoretical frameworks related to indirect effects, the following mediated relationships were hypothesized:

H5: Higher ICT development indirectly increases the likelihood of ChatGPT use through enhanced access, more favorable linguistic support, and stronger institutional support.

- **H5a:** Higher ICT development indirectly increases the likelihood of ChatGPT use through improved functional access.
- **H5b:** Higher ICT development indirectly increases the likelihood of ChatGPT use through more favorable linguistic support.
- **H5c:** Higher ICT development indirectly increases the likelihood of ChatGPT use through stronger institutional support.

National digital development may influence ChatGPT use not only indirectly, but also directly. This effect may be shaped by systemic factors such as national innovation strategies, societal acceptance of technology, and educational policy priorities (Wesna et al., 2025). Given that digital development may independently affect tool usage, we also formulated a hypothesis regarding a direct effect.

H6: Higher ICT development has a positive direct effect on ChatGPT use, which is only partially mediated by access-related, institutional, and linguistic factors.

- **H6a:** Higher ICT development has a positive direct effect on ChatGPT use even when controlling for ACCESS.
- **H6b:** Higher ICT development has a positive direct effect on ChatGPT use even when controlling for LANGUAGE_SUPPORT.
- **H6c:** Higher ICT development has a positive direct effect on ChatGPT use even when controlling for INSTITUTIONAL_SUPPORT.

4. Results

According to the results of the structural model, ACCESS has a significant medium-sized effect on ChatGPT use ($\beta = .341$, $t = 11.7$, $p < .001$, $R^2 = .001$). The direct effect of IDI is weak but statistically significant ($\beta = -.017$, $t = 5.3$, $p < .001$). Among the mediators, LANGUAGE_SUPPORT ($\beta = .006$, $p = .366$) and SUPPORT ($\beta = .002$, $p = .690$) did not show significant associations, whereas the ACCESS $\rightarrow$ LANGUAGE_SUPPORT path was strong and significant ($\beta = .498$, $t = 34.2$, $p < .001$, $R^2 = .015$). The indirect effect of IDI was substantial (H5) and contributed meaningfully to the overall explanatory power of the model (H6).

4.1. Demographic characteristics of respondents

The key demographic, methodological, and outcome variables of the study are summarized in the tables below; all detailed values are directly accessible (Tables 1 and 2).

4.2. Independent variables of ChatGPT usage: country and digital development

To examine the impact of student background variables and national context on ChatGPT use, we conducted a binary logistic regression including three predictors: citizenship, country of study, and the ICT Development Index (IDI). The dependent variable was binary (user vs. non-user). The results of the analysis are presented in Table 3. According to the findings, country of study had a statistically significant but moderate effect on the likelihood of ChatGPT use ($B = -.003$; $p = .007$), whereas citizenship was not a significant predictor ($p = .154$). This suggests that the location of education may play a more important role in shaping access to AI tools than students' country of origin. The strongest effect was observed for digital development: higher IDI scores significantly decreased the likelihood of *not* using ChatGPT ($B = -.967$; $p < .001$; $\text{Exp}(B) = .380$). The negative coefficient reflects the model's coding, in which "non-use" was the reference category – thus, a higher IDI value indicates an increased likelihood of ChatGPT use.

However, the explanatory power of the models remained low. The model including citizenship and country of study showed weak fit

Table 1
Demographic Characteristics of the sample.

Variable	Category	Valid Percent (%)
Gender	Male	41.01
	Female	57.66
	Other	0.41
Student status	Full-time	85.37
	Part-time	14.63
Level of study	Bachelor	83.55
	Master	12.36
	PhD	4.09
Main field of study	Arts and Humanities	11.24
	Social Sciences	39.43
	Applied Sciences	37.55
	Natural and Life Sciences	11.78
Self-assessed economic status	Significantly below average	6.21
	Below average	2.10
	Average	55.24
	Above average	16.01
	Significantly above average	2.44

Source: own editing

Table 2
Descriptive statistics – age.

Statistic	Value
Valid N	2.189
Mean	23.09
Standard Deviation	6.586

Source: own editing based on SPSS output

Table 3
Logistic regression coefficients for ChatGPT usage.

Variable	B	S.E.	Wald	df	Sig.	Exp(B)
Citizenship	.002	.001	2.032	1	.154	1.002
Country of study	-.003	.001	7.332	1	.007	.997
ICT Development Index	-.967	.016	3732.686	1	<.001	.380
Constant	-.855	.033	655.960	1	<.001	.425

Source: own editing based on SPSS output

(Nagelkerke $R^2 = .001$), and the model extended with digital development also performed poorly (Nagelkerke $R^2 = .002$; Hosmer–Lemeshow tests: $p < .001$ in both cases). These results suggest that additional factors – such as institutional and individual-level variables – may also influence ChatGPT use.

The findings support **Hypothesis 2**, indicating that the country of study significantly affects ChatGPT use, and **Hypothesis 3**, which posits that national digital development is also a relevant factor. However, **Hypothesis 1** – concerning the impact of citizenship – was not supported.

4.3. Impact of demographic and background factors on ChatGPT usage

To further explore the background factors associated with ChatGPT use, a binary logistic regression was conducted (ENTER method), including all relevant demographic and contextual variables. The dependent variable was based on whether the respondent had ever used ChatGPT (1 = yes, 0 = no).

The results, presented in Table 4, show that several background variables had a significant impact on the likelihood of ChatGPT use. Higher usage rates were observed among students who:

- studied in countries with higher levels of digital development (IDI),
- identified as male,
- reported above-average financial status,
- were enrolled in master's or doctoral programs,
- were studying in non-humanities fields, and
- participated in online or hybrid forms of education.

In contrast, lower usage rates were observed among students in traditional, in-person education formats, those with lower income levels, and those enrolled in humanities or applied sciences programs.

It is worth noting that even in countries with very high IDI scores, there were substantial differences in usage frequency. This suggests that structural access alone does not guarantee the active use of the tool.

As the study is cross-sectional in nature, the variables included in the model are treated as independent variables, without drawing any causal inferences. The strongest effects were observed for digital development (IDI) and learning format: students engaged in online learning were over 60 % more likely to use ChatGPT, and the likelihood of use significantly increased in countries with high or very high IDI levels. Socioeconomic background and field of study also showed consistent patterns: students with lower income levels and those enrolled in humanities programs were less likely to take advantage of AI tools. Although the model's explanatory power was moderate (Nagelkerke $R^2 = .041$), the results clearly indicate that student background and contextual factors play a key role in the adoption of AI tools. However, the overall model fit was

Table 4

Significant independent variables of ChatGPT use (binary logistic regression, ENTER method).

Independent Variables	B	p-value	Exp(B)	Interpretation
ICT Development Index (IDI)	-.967	<.001	.380	Higher IDI → greater likelihood of usage
Gender (Male)	-.406	.044	.666	Males more likely to use ChatGPT
Gender = Female (vs. Male)	-.142	.012	.868	Females less likely
Economic Status (Above Avg.)	.300	.024	1.350	Higher SES increases likelihood
Economic Status (Below Avg.)	.392	.002	1.481	Lower SES decreases likelihood
Level of Study (Master)	.440	<.001	1.552	Master's students more likely
Level of Study (PhD)	.310	.010	1.364	PhD students more likely
Field = Humanities	.367	.001	1.444	Humanities students use it less frequently
Field = Applied Sciences	-.367	<.001	.693	Applied sciences students less likely to use
Learning Modality = Traditional	.335	.004	1.398	Traditional learners less likely
Learning = Online	.470	<.001	1.600	Online learners more likely
Learning = Hybrid	.404	<.001	1.498	Hybrid learners more likely
IDI (4)	3.138	<.001	23.068	High IDI strongly increases usage
IDI (9)	3.182	<.001	24.088	Very high IDI → very strong effect
IDI (10)	2.723	<.001	15.219	Very high IDI → very strong effect

Source: own editing based on SPSS output

weak (Hosmer–Lemeshow test: $p < .001$), suggesting that other factors – such as motivation, prior experience, or institutional culture – may also influence students' willingness to use these tools.

Among the sub-hypotheses related to H4, statistically significant

support was found for **H4a–H4d** (gender, financial status, field of study, and learning modality). In contrast, **no significant effects** were observed for **age (H4e)**, **student status (H4f)**, **level of study (H4g)**, and **employment status (H4h)**.

4.4. PLS-SEM mediation model

To explore the underlying mechanisms of ChatGPT use, we applied a PLS-SEM model that examined both the direct and indirect effects of digital development (IDI) through three mediating variables: functional access (ACCESS), linguistic support (LANGUAGE_SUPPORT), and institutional support (SUPPORT) (Fig. 3). Minor discrepancies observed in the text, figure, and tables are due to rounding differences specific to SmartPLS.

The reliability and validity of the measurement model were acceptable (Tables 5 and 6). All constructs met the established criteria (Cronbach's $\alpha > .70$; HTMT $< .85$; VIF < 5). Although the AVE value for the ACCESS construct was slightly low (.49), it is still considered acceptable in the context of exploratory research.

The structural model confirmed partial mediation. The IDI had a small but statistically significant direct effect on ChatGPT use ($\beta =$

Table 5

Composition of constructs.

Latent Variable	Survey Items	Description
USED_ChartGPT	Q13	ChatGPT usage (yes/no)
ACCESS	Q15, Q18a, Q18b, Q18g, Q18h, Q18i, Q18k	Intensity, purpose, and functionalities of use
LANGUAGE_SUPPORT	Q28d, Q28f	Native language competencies, interpersonal communication
SUPPORT	Q21c, Q23a–c	Ethical regulation, instructor consultation, feedback
IDI	–	ICT Development Index (ITU, 2023)

Source: own editing based on SmartPLS output

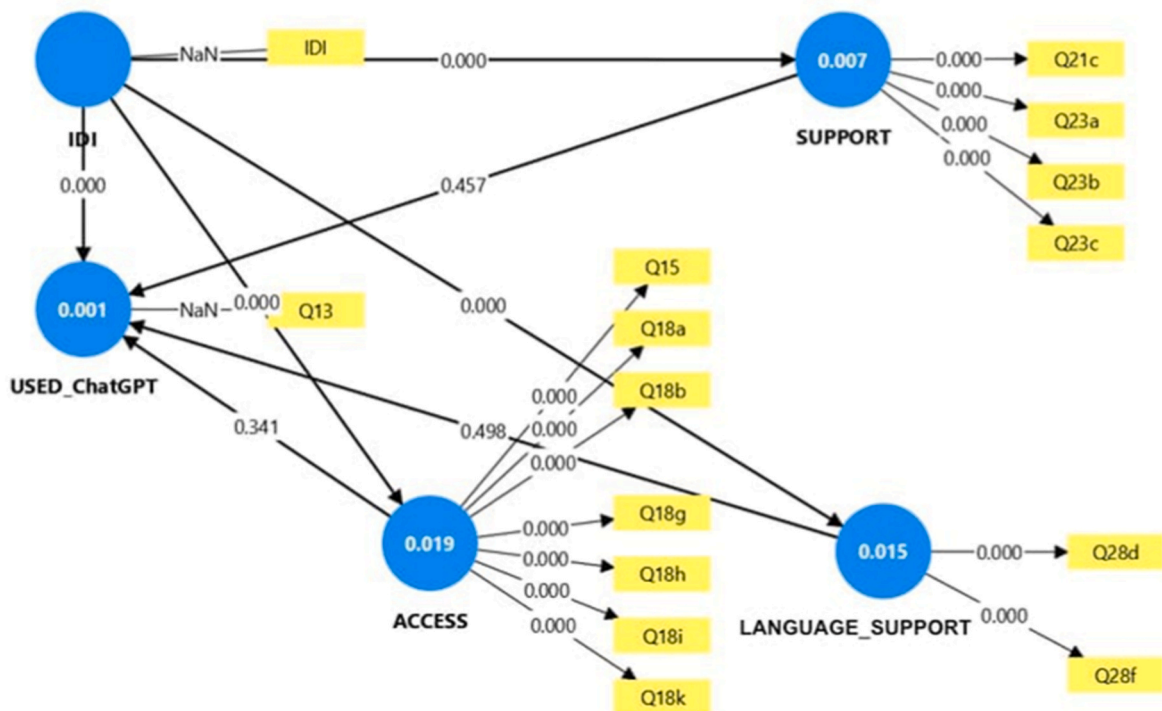


Fig. 3. Mediation Model. Source: SmartPLS output.

Table 6
Reliability and convergent validity indicators.

Latent Variable	Cronbach's Alpha	Composite Reliability	AVE	Remarks
ACCESS	.83	.87	.49	AVE is slightly low, can be improved by removing Q18h or Q18k
LANGUAGE_SUPPORT	.74	.87	.77	High convergent validity
SUPPORT	.75	.83	.57	All item loadings are adequate
IDI	1.00	1.00	1.00	Single-item formative construct
USED_ChatsGPT	1.00	1.00	1.00	Single-item binary variable

Source: own editing based on SmartPLS output

-.017; $p < .001$), along with a strong indirect effect mediated by functional access ($\beta = .341$; $p < .001$). This indicates that students who actively engage with various ChatGPT functions are more likely to use the tool on a regular basis. IDI also significantly reduced linguistic and institutional barriers; however, these factors did not show a direct association with usage (Table 7).

Overall, both **Hypothesis 5** (indirect effect) and **Hypothesis 6** (remaining direct effect) were supported. While IDI influences ChatGPT usage directly, this effect is further amplified through functional access – highlighting that both structural and user-level factors contribute to AI adoption in higher education. To provide a clear overview of the statistical findings, the table below outlines each hypothesis, its formulation, the corresponding test results, and whether it was supported. This format allows for transparent evaluation of the empirical evidence in relation to the theoretical expectations (Table 8). For Hypotheses 1 and 2, the effects were examined through a preliminary analysis independent of the PLS-SEM model, as these variables were not included in the structural path model.

4.5. Cluster profiles

To explore patterns of ChatGPT usage, we conducted a TwoStep cluster analysis based on four key variables: functional access (ACCESS), institutional support (SUPPORT), linguistic support (LANGUAGE_SUPPORT), and national digital development (IDI). The procedure identified three clearly interpretable clusters with moderate solution quality (Silhouette coefficient = .35). The main characteristics of the clusters are summarized in Table 9.

- **Cluster 1 – Digital Pioneers (35.3 %):** This cluster includes students from moderately developed digital environments (IDI = 65.0) who make intensive use of ChatGPT (ACCESS = 5.00) and report high levels of linguistic support (5.00). Although institutional support is low (2.50), their digital competence likely developed independently.

Table 7
Estimation of structural paths.

Path	Beta	t	p	R ²	Interpretation
IDI → ACCESS	-.137	21,9	<.001	.019	Higher digital development is associated with fewer functional barriers
IDI → LANGUAGE_SUPPORT	-.124	19,6	<.001	.015	Linguistic and cultural barriers are lower in more developed countries
IDI → SUPPORT	-.082	14,5	<.001	.007	Need for institutional support decreases in more advanced environments
IDI → USED_ChatsGPT	-.017	5,3	<.001	.001	Direct effect remains significant even with mediators
ACCESS → LANGUAGE_SUPPORT	.498	34,2	<.001	.015	Functional diversity promotes linguistic advantages
ACCESS → USED_ChatsGPT	.341	11,7	<.001	.001	Broader use of ChatGPT features strongly predicts usage
LANGUAGE_SUPPORT → USED_ChatsGPT	.006	.9	.366	.001	Non-significant effect
SUPPORT → USED_ChatsGPT	.002	.4	.690	.001	Non-significant effect

Source: own editing based on SmartPLS output

Table 8
Hypothesis testing outcomes.

Hypothesis	Statement	Statistical Result	Supported?
H1	Citizenship → ChatGPT use	Not significant	Not Supported
H2	Country of study → ChatGPT use	Significant	Supported
H3	IDI → ChatGPT use (direct)	Significant	Supported
H4a	Gender → ChatGPT use	Significant	Supported
H4b	Financial status → ChatGPT use	Significant	Supported
H4c	Field of study → ChatGPT use	Significant	Supported
H4d	Mode of study → ChatGPT use	Significant	Supported
H4e	Age → ChatGPT use	Not significant	Not Supported
H4f	Student status → ChatGPT use	Not significant	Not Supported
H4g	Level of study → ChatGPT use	Not significant	Not Supported
H4h	Employment status → ChatGPT use	Not significant	Not Supported
H5a	IDI → ACCESS → ChatGPT use	Significant indirect effect	Supported
H5b	IDI → LANGUAGE_SUPPORT → ChatGPT use	Not significant	Not Supported
H5c	IDI → SUPPORT → ChatGPT use	Not significant	Not Supported
H6a	IDI → ChatGPT use (direct effect retained)	Significant	Supported
H6b	ACCESS → ChatGPT use	Significant	Supported
H6c	LANGUAGE_SUPPORT → ChatGPT use; SUPPORT → ChatGPT use	Not significant	Not Supported

Source: own editing

- **Cluster 2 – Institutional Catch-Ups (6.9 %):** This group consists of students from countries with low digital development (IDI = 25.8). They report moderate ChatGPT usage (ACCESS = 3.57), yet benefit from high levels of institutional (5.00) and linguistic support (4.00), possibly due to active faculty engagement.
- **Cluster 3 – Underutilized Digital Elite (57.8 %):** This cluster comprises students from highly developed digital countries (IDI = 96.9) who nevertheless make minimal use of ChatGPT (ACCESS = 1.29) and report low linguistic (1.00) and institutional support (3.50). Despite strong infrastructure, usage appears to be hindered by a lack of supportive academic context.

In terms of gender distribution, the Digital Pioneers cluster is balanced (51.3 % male, 47.6 % female), whereas the other two clusters are female-dominated, particularly the Underutilized Digital Elite group (57.8 % female). Across all clusters, the majority of students fall within the 18–23 age range. However, the Institutional Catch-Ups cluster has the highest proportion of students over 24 years old (33.6 %), suggesting a more mature and experienced learner profile. Regarding educational level, the Digital Pioneers and Underutilized Digital Elite clusters are predominantly composed of undergraduate (BA/BSc) students (66.5 % and 64.0 %, respectively), while the Institutional Catch-Ups group displays greater diversity: 34.0 % are enrolled in Master's programs and 1.7

Table 9
Cluster profiles – comparative summary.

Variable	Digital Pioneers	Institutional Catch-Ups	Underutilized Digital Elite
IDI (ICT Development Index)	65.0	25.8	96.9
ChatGPT Usage (ACCESS)	5.00	3.57	1.29
Institutional Support	2.50	5.00	3.50
Linguistic Support	5.00	4.00	1.00
Sample Size	4169 (35.3 %)	817 (6.9 %)	6821 (57.8 %)
Female	47.6 %	54.7 %	57.8 %
Male	51.3 %	44.2 %	4.9 %
% Other/No response (Gender)	1.1 %	1.1 %	1.3 %
Age 18	1.3 %	4.0 %	9.6 %
Age 19	13.3 %	6.5 %	15.7 %
Age 20	14.8 %	12.3 %	17.7 %
Age 21	19.1 %	19.1 %	14.6 %
Age 22	12.9 %	12.3 %	7.7 %
Age 23	8.9 %	1.2 %	7.7 %
Age 24+	2.7 %	33.6 %	27.0 %
BSc/BA students	66.5 %	43.8 %	6.4 %
MSc/MA students	21.8 %	34.0 %	26.8 %
PhD students	5.5 %	1.7 %	6.0 %
Other/No response (Education)	6.2 %	11.5 %	6.8 %
Full-time students	71.3 %	59.5 %	65.8 %
Part-time students	24.6 %	33.2 %	29.1 %
Other/No response (Study mode)	4.1 %	7.3 %	5.1 %

Source: own editing based on SmartPLS output

% in doctoral studies. In terms of study mode, full-time students are the majority in all clusters; however, the Institutional Catch-Ups cluster shows the highest proportion of part-time learners (33.2 %), potentially indicating parallel employment or alternative learning trajectories.

5. Discussion

This study examined the integration of generative artificial intelligence (AI) in higher education through a multi-level framework of digital inequality. The findings indicate that ChatGPT use is shaped not only by national technological development, as measured by the ICT Development Index (IDI), but also by institutional conditions and individual access-related factors. Usage was more prevalent in countries with higher IDI scores, highlighting the enabling role of a developed digital ecosystem (Jin et al., 2025). However, infrastructure alone is insufficient; institutional guidance and educational policy support are also critical (Wesna et al., 2025; Bukhari, 2025).

Functional access, including digital device use, learning modality, and digital competence, proved a stronger predictor than IDI alone. Students engaged in online or hybrid learning demonstrated more intentional AI use, whereas technological uncertainty was more common in face-to-face settings (Zhang, 2024). Ethical and confident use was particularly evident in institutions with clear policies and active faculty support (Ful Haq et al., 2025). Disciplinary and cultural variations also emerged: students in STEM and business programs were generally more receptive, while those in the humanities and social sciences expressed greater skepticism, particularly regarding originality and academic integrity (Chan & Colloton, 2024). Gender and language-related differences were also notable: male students tended to be more open, while female students reported more ethical concerns and language-related uncertainty, especially regarding English-language prompting (Sharma & Ha, 2025). Although ChatGPT is technically multilingual, language barriers may still limit its practical use (Rochmah, 2025). The mediation model showed that IDI primarily influences AI use indirectly through functional access, although a direct effect also remained significant. The negative coefficient in the model indicates that higher IDI scores are

associated with greater likelihood of use. Cluster analysis identified three student profiles. Digital Pioneers were from countries with moderate IDI levels but demonstrated high usage and digital competence, often without institutional support. In contrast, the Underutilized Digital Elite, students from countries with the highest IDI, reported low usage, suggesting that infrastructure alone does not ensure adoption (Duong et al., 2024). The third group, Institutional Catch-Ups, consisted of students from low-IDI contexts who nonetheless benefited from strong institutional and language support, likely due to proactive faculty engagement (Ful Haq et al., 2025). This group had a notably high proportion of part-time students, underscoring the importance of balancing learning and work in the context of AI use (Apatha et al., 2025).

Although the regression models revealed significant relationships, their explanatory power remained low (Nagelkerke $R^2 = .001-.041$), suggesting that ChatGPT use is more strongly influenced by individual factors such as motivation, self-efficacy, and perceived usefulness, as well as by institutional attitudes (Alamäki et al., 2024). Another limiting factor is ecological bias: national IDI scores do not necessarily reflect the realities of individual students or institutions (Helsper, 2021). This is supported by the cluster analysis, which found low usage even in several high-IDI countries (Ravšelj et al., 2024). These findings highlight the need to examine psychological and learning-related variables, such as the TAM model, digital self-efficacy, and perceived usefulness, alongside structural indicators (Banihashem et al., 2025). The results point to a multi-level adaptation model: at the macro level, national digital infrastructure (IDI) provides the structural foundation; functional access, including device use, learning modality, and digital skills, serves as a mediating mechanism. At the meso level, institutional context, faculty attitudes, and support structures shape adoption, while at the micro level, individual attitudes, digital self-efficacy, and linguistic or cultural capital determine actual use (Van Dijk, 2005). This framework helps interpret how various factors interact in the adoption of generative AI. The study was based on a convenience sample and cross-sectional design, meaning the results are not representative and do not allow for causal inference (Chiu, 2024). Unequal sample sizes, lack of institutional-level data, and the aggregated nature of IDI may have introduced bias (Ravšelj et al., 2024). While the measurement model showed overall reliability, some constructs – especially functional access – require improvement in AVE values. Future research should refine measurement tools (e.g., prompt design skills, ethical sensitivity), include institutional-level variables, and apply multi-level modeling (Hair et al., 2021). The integration of generative AI in higher education is a complex, multi-dimensional phenomenon that requires a holistic approach that considers inequalities in access, support, and digital competence (Apatha et al., 2025).

6. Conclusion

The findings of this study suggest that integrating generative artificial intelligence (AI) in higher education requires systemic adaptation rather than mere technological advancement. While a higher ICT Development Index (IDI) continues to increase the likelihood of AI use (Jin et al., 2025), actual AI tool adoption is primarily shaped by institutional environments and individual-level factors. Therefore, infrastructure development alone is not sufficient; meaningful progress depends on clear institutional guidance, educational policy support, and targeted development of digital competencies (Wesna et al., 2025). Results related to functional access and digital skills underscore that higher education institutions must provide not only technological resources but also pedagogical and supportive environments that promote conscious, ethical, and responsible use of AI (Anders & Speltz, 2025). Cultural, disciplinary, gender, and linguistic differences must also be addressed, as they influence not just access but also student experience and equity (Fan, 2025).

The study further highlights that national-level IDI scores do not always reflect institutional or regional realities, underscoring the risk of

ecological bias and the need for multi-level statistical approaches (Helsper, 2021). Differences across clusters suggest that effective AI adaptation can occur even in less favorable macro-contexts if strong institutional support and digital awareness are present (Apata et al., 2025). Overall, the study emphasizes that advancing the use of generative AI in higher education is a strategic, pedagogical, and societal task that extends far beyond technological infrastructure. Future research should examine institutional-level integration, further refine measurement tools (such as those assessing AI literacy and prompt design skills), and explore the dynamic interplay of macro-, meso-, and micro-level factors (Shi et al., 2025).

CRediT authorship contribution statement

Andrea Bencsik: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Dr. **Szonja Jenei:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization, Dr.

AI usage statement

During the preparation of this work, the authors used ChatGPT solely to improve the language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the submitted article.

Ethical statement

The present study is based on a publicly accessible dataset collected during an international research project conducted between early October 2023 and the end of February 2024.

The original research was initiated by the CovidSocLab research group and coordinated by the Faculty of Public Administration, University of Ljubljana, under the leadership of Prof. Dr. Aleksander Aristovnik. The project was implemented through a global collaboration involving researchers from numerous countries.

The questionnaire-based data collection complied fully with international ethical standards and the principles of the Declaration of Helsinki, specifically:

- participation was anonymous,
- completion of the questionnaire was entirely voluntary,
- all data were used in an aggregated form only,
- data processing adhered to all relevant provisions of the GDPR,
- the questionnaire contained no personal identifiers,
- only opinions and self-reported experiences were collected.

The original survey instrument and dataset are publicly available at the following link:

Ravšelj D, Aristovnik A, Keržič D, Tomažević N, Umek L, Brezovar N et al. (2025).

Higher education students' early perceptions of ChatGPT: Global survey data.

Mendeley Data. <https://doi.org/10.17632/yng9nsn6kn>.

One of the authors of the present article, Prof. Dr. Andrea Bencsik, joined the research project after the data collection phase and contributed to the preparation of the first scientific publication. That publication involved nearly 150 international researchers, highlighting the highly collaborative and global nature of the project. The entire publication process fully complied with all relevant ethical requirements.

Since then, the open-access dataset has been used in several additional international publications, such as: Ravšelj D, Keržič D, Tomažević N, Umek L, Brezovar N, Iahad N. A. et al. (2025). *Higher education*

students' perceptions of ChatGPT: A global study of early reactions. PLoS ONE, 20(2): e0315011. <https://doi.org/10.1371/journal.pone.0315011>.

The present study builds on the aforementioned publicly available dataset. No new data were collected. As the original research fully adhered to all ethical standards, and the current analysis is limited to processing anonymized, aggregated data, no additional ethical approval was required for this study.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The anonymized datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

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